CUADERNO DE EXAMEN

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CARRERA : Física CURSO : FS0430

☐ BACHILLERATO ☐ LICENCIATURA ☐ MAESTRIA ☐ TECNICO

EXAMEN : ☐ I PARCIAL ☐ II PARCIAL ☐ FINAL ☐ SUFICIENCIA

RB	10
1	15
2	24
3	11

FECHA : Sept 26, '25 PUNTOS OBTENIDOS :

NOTA : 60

Respuesta breve.

① En A, porque en este caso $V(x) \propto E(x)$

② Será efectivo en (c).

Para (b), el el $\vec{E}$ enc. por Gauss es $\frac{q_{enc}}{\varepsilon_0 A}$. Afuera, depende

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de la existencia de otras cargas.

En (c), el $\vec{E}_{enc} = 0$, porque $q_{enc} = 0$. No depende de si hay otras cargas afuera.

(3)

$+Q \quad r \quad -q$

La que tiene mayor energía potencial eléctrica es ~~$-q$~~

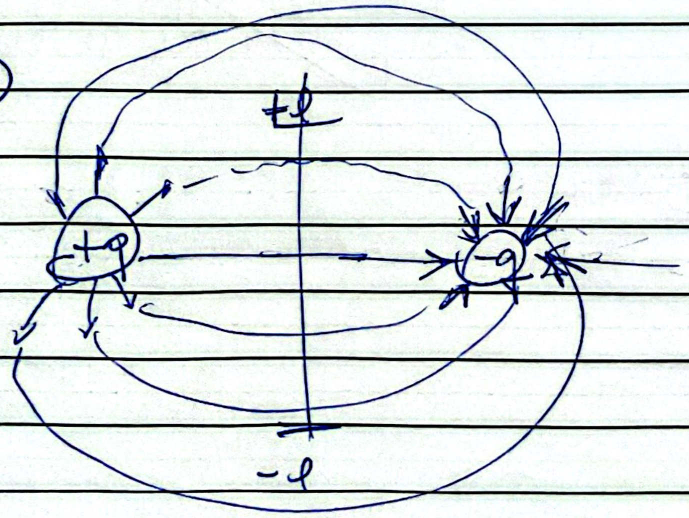
al revés porque la carga fuente es $+$, por lo que no requiere trabajo para acercarlas.

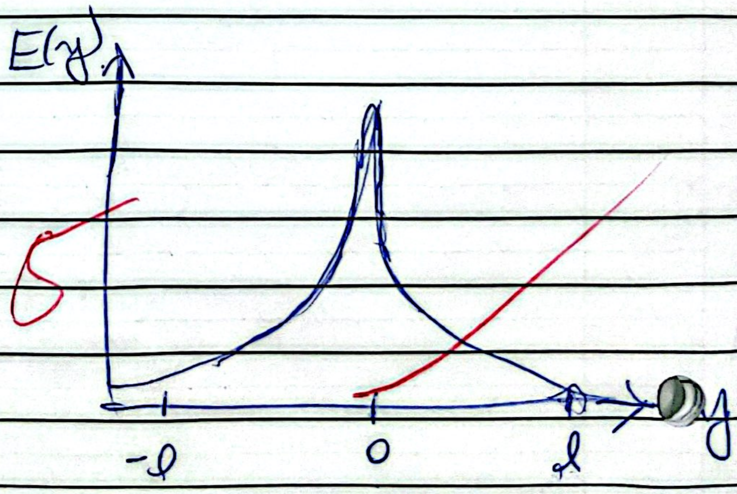
La que tiene mayor Potencial eléctrico es $+q$

Porque

$$\cancel{\frac{k_e(+q)(+Q)}{r} > \frac{k_e(-q)(+Q)}{r}}$$

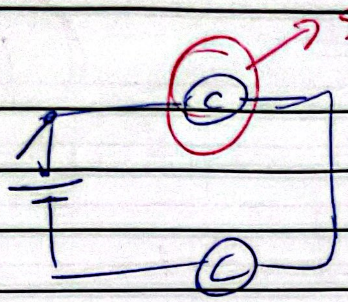
(4)





$$E = E_+ + E_- = \frac{k_e q^+}{r^2} + \frac{k_e q^-}{r^2}$$

⑤



símbolos no son correctos

Al cerrar el switch las cargas fluyen. La batería genera la corriente ¿cómo?

Las cargas vienen justamente de la batería.

Problema #1

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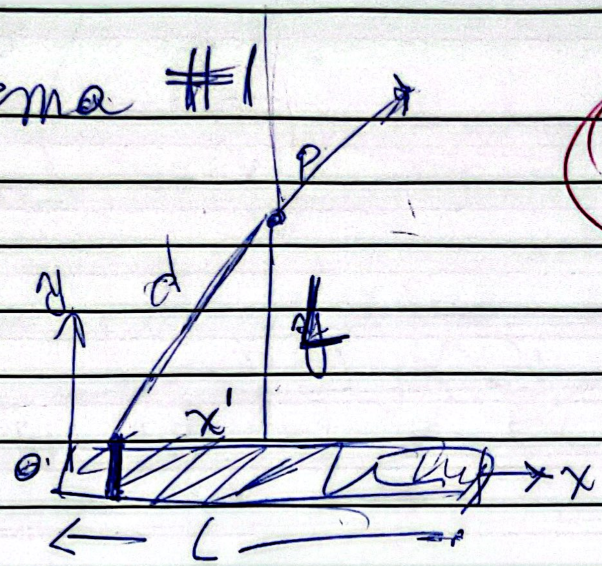


$\lambda = \lambda_0 + \lambda_1 x$

$d^2 = x'^2 + y^2$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda}{d^2}\, dx' \,\hat{r}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda_0 + \lambda_1 x'}{x'^2 + L^2}\, dx' \left\langle \frac{L}{d}, \frac{x'}{d} \right\rangle$$

$X \neq x'$

$$E_x = \frac{1}{4\pi\varepsilon_0} \int_0^L \frac{(\lambda_0 + \lambda_1 x') \cdot L\, dx}{(x'^2 + L^2)^{3/2}}$$

$x' = \frac{L}{2} - X$

$$E_y = \frac{1}{4\pi\varepsilon_0} \int_0^L \frac{(\lambda_0 + \lambda_1 x') \cdot x'\, dx}{(x'^2 + L^2)^{3/2}}$$

$$E_x = K_e \lambda_0 L \int_0^L \frac{dx}{(x'^2 + L^2)^{3/2}} + K_e \lambda_1 L \int_0^L \frac{x'\, dx}{(x'^2 + L^2)^{3/2}}$$

$$E_x = +\frac{K_e \lambda_0 L}{L^2}\left(\frac{x'}{\sqrt{x'^2+L^2}}\right)_0^L$$

$$- K_e \lambda_1 L \left(\frac{1}{\sqrt{x'^2+L^2}}\right)_0^L$$

$$\vec{E}_x = \frac{K_e \lambda_0}{L}\left(\frac{\cancel{L}}{\cancel{L}\sqrt{2}}\right)$$

$$- K_e \lambda_1 L \left(\frac{1}{L\sqrt{2}}\right) \quad \frac{1}{L}$$

Para $\bar{E}_y$:

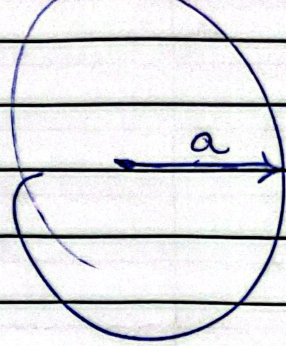
$$\bar{E}_y = K_e \lambda_0 \int_0^L \frac{x'\,dx}{(x'^2+L^2)^{3/2}} + K_e \lambda_1 \int_0^L \frac{x'^2\,dx}{(x'^2+L^2)^{3/2}}$$

$$= -K_e \lambda_0 \left(\frac{1}{\sqrt{x'^2+L^2}}\right)_0^L + K_e \lambda_1 \left[\ln\left(x+\sqrt{L^2+x^2}\right) - \frac{x}{\sqrt{L^2+x^2}}\right]_0^L$$

$$= -K_e \lambda_0 \left(\frac{1}{L\sqrt{2}} - \frac{1}{L}\right) + K_e \lambda_1 \left[\ln(L+L\sqrt{2}) - \frac{1}{\sqrt{2}} - \ln(L)\right]$$

Problema #2

$$\rho = a^2 - r^2$$



$$\boxed{\text{Para } r < a:}$$

$$\int E_1\, dA = \frac{q}{\varepsilon_0}$$

$$E_1\, 4\pi r^2 = \frac{1}{\varepsilon_0}\int_V \rho\, dV$$

$$E_1 = \frac{1}{\cancel{4\pi r^2}\varepsilon_0}\int_0^r (a^2 - r^2)\,\cancel{4\pi r^2}\, dr$$

$$E_1 = \frac{1}{r^2\varepsilon_0}\left(\frac{a^2 r^3}{3} - \frac{r^5}{5}\right)\Bigg|_0^r$$

$$\boxed{\vec{E}_1 = \frac{1}{\varepsilon_0}\left(\frac{a^2 r}{3} - \frac{r^3}{5}\right)\hat{r}}$$

$$V_1 = -\int_0^{ar} \bar{E}_1\, d\bar{r} + C_1$$

$$V_1 = \frac{1}{\varepsilon_0}\int\left(\frac{a^2 r}{3} - \frac{r^3}{5}\right) dr + C_1$$

$$V_1 = \frac{1}{\varepsilon_0}\left(\frac{a^2 r^2}{6} - \frac{r^4}{20}\right) + C_1$$

si $r = a$

$$V_r = \frac{1}{\varepsilon_0}\left(\frac{a^4}{6} - \frac{a^4}{20}\right) + C_1$$

$$\rightarrow C_1 = V_r + \frac{1}{\varepsilon_0}\left(\frac{3a^4 - 10a^4}{60}\right)$$

$$\rightarrow C_1 = V_r - \frac{7a^4}{60\varepsilon_0}$$

$$V_1 = \frac{1}{\varepsilon_0}\left(\frac{a^2 r^2}{6} - \frac{r^4}{20}\right) + V_r - \frac{7a^4}{60\varepsilon_0}.$$

Para $a < r < \infty$.

$$\int E_2 \, dA = \frac{q}{\varepsilon_0}$$

$$E_2 = \frac{1}{\cancel{4\pi r^2}\varepsilon_0} \int_0^a (a^2 - r^2)\, \cancel{4\pi} r^2 \, dr$$

$$E_2 = \frac{1}{r^2\varepsilon_0}\left(\frac{a^2r^3}{3} - \frac{r^5}{5}\right)_0^a$$

$$E_2 = \frac{1}{r^2\varepsilon_0}\left(\frac{a^5}{3} - \frac{a^5}{5}\right)$$

$$E_2 = \frac{1}{r^2\varepsilon_0}\left(\frac{5a^5 - 3a^5}{15}\right)$$

$$\boxed{\vec{E}_2 = \frac{2}{15}\frac{a^5}{r^2\varepsilon_0}\hat{r}}$$

$$V_2 = -\int_a^\infty \bar{E}_2\, d\bar{r} + C_2$$

$\rightsquigarrow r > a$

$$V_2 = -\frac{2a^5}{15\varepsilon_0}\int \frac{1}{r^2}\,dr + C_2$$

$$V_2 = \frac{2}{15}\frac{a^5}{\varepsilon_0}\left(\frac{1}{r}\right)_a^\infty + C_2$$

$$V_2 = \frac{2}{15}\frac{a^5}{\varepsilon_0}\left(0 - \frac{1}{a}\right) + C_2$$

$$V_2 = -\frac{2}{15}\frac{a^{4\,5}}{\varepsilon_0 r} + C_2$$

Si $r = a$

$$V_2 = V_r$$

$$-\frac{2}{15}\frac{a^{\cancel{4}5}}{\varepsilon_0 r} + C_2 = V_r$$

$$C_2 = V_r + \frac{2}{15}\frac{a^4}{\varepsilon_0}.$$

$$V_2 = \frac{4}{45} V_r \quad (?)$$

La integral para el potencial V_2 era de r hasta infinito. Muy bien el resto!

Problema #3

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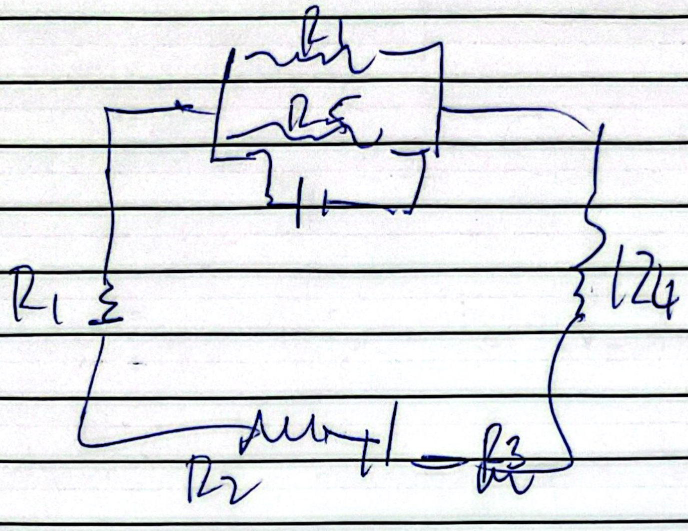
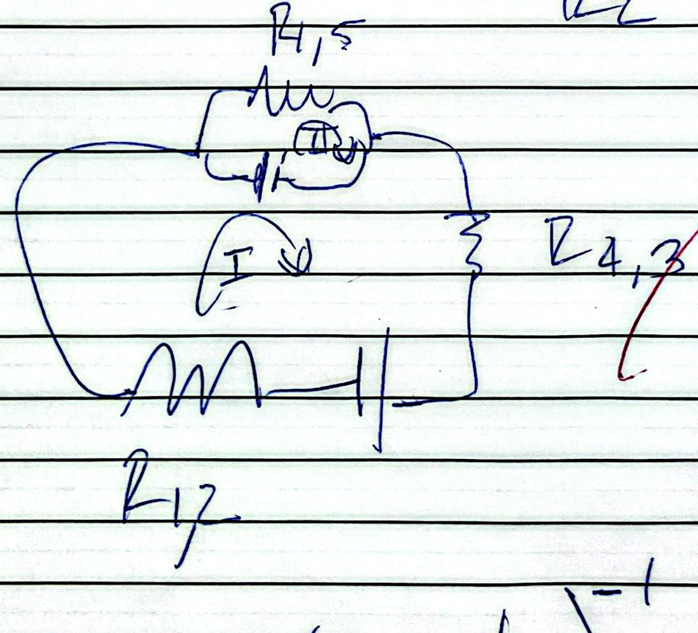


$R_1 \parallel R_5$

$R_4 \leftrightarrow R_3$

$R_2 \leftrightarrow R_1$



$$R_{1,5} = \left(\frac{1}{R_1} + \frac{1}{R_5}\right)^{-1}$$

$$R_{1,2} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{4,3} = \frac{R_4 R_3}{R_4 + R_3}$$

$$\varepsilon_1 + \varepsilon_2 - i_1 R_{1,2} - i_1 R_{4,3} = 0$$

$$\varepsilon_1 - i_2 R_{1,5} = 0$$

$$i_1 = i_2 + i_3$$

$$i_1 = \frac{\varepsilon_2}{R_{1,2} + R_{4,3}}$$

$$i_1 = \frac{15\,V}{\frac{R_1 R_2}{R_1 + R_2} + \frac{R_4 R_3}{R_4 + R_3}}$$ ¿Unidades!?

$$i_1 = \frac{15}{\frac{500 \cdot 10}{500 + 10} + \frac{50 \cdot 100}{50 + 100}}$$

$$= \frac{15}{\left(\frac{500}{51} + \frac{500}{15}\right)} =$$